

ST BENEDICT'S COLLEGE



SUBJECT
GRADE
EXAMINER
NAME
TEACHER

Mathematics Paper 1

11

Mr Benecke

DATE

MARKS

MODERATORS

DURATION

14+17 February 2017

70

Mrs Povall + Grade 11 teachers

80min

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Equations and inequalities	36	
2	Simultaneous	9	
3	Nature of Roots	6	
4	Completing the Square	11	
5	Word Problem	8	
TOTAL		70	

QUESTION 1**36 MARKS**a) Solve for x :

i) $x^2 - 11x - 12 = 0$ (4)

ii) $(x + 1)(x - 3) = 8$ (6)

iii) $\frac{1}{x^2-2x} - \frac{8}{3x^2-5x-2} = \frac{4}{3x^2+x}$ (8)

iv) $x = \sqrt{x+3} - 1$ (6)

v) $x^2 + 4x + \frac{12}{x^2+4x} = -7$ (8)

vi) $x^2 + 5x - 6 < 0$ (4)

QUESTION 2**9 MARKS**

Solve for x and y in the following equations:

$$2y - x = 1 \text{ and } y - 2x^2 + 3x + 5 = 0 \quad (9)$$

QUESTION 3**11 MARKS**

a) Given: $x^2 + 6x + 8 = 0$

Solve for x by completing the square

(4)

b) Given: $f(x) = 2x^2 - 5x - 12$

(7)

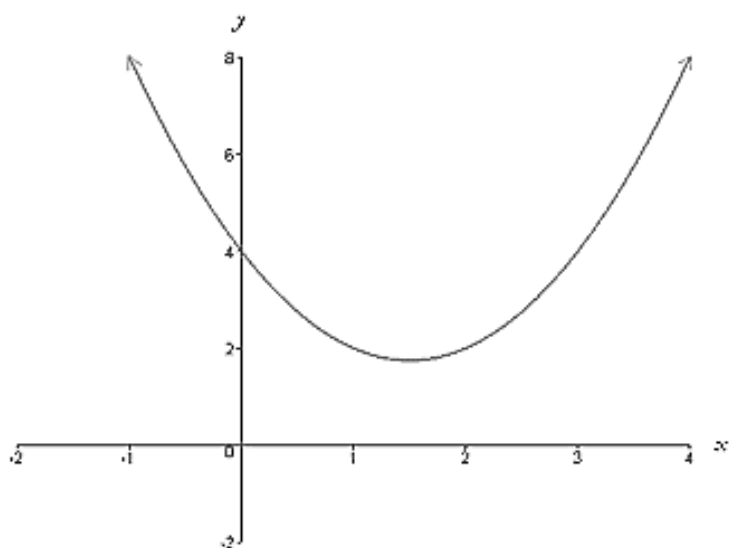
Determine the minimum value of $f(x)$ and the corresponding x value that generates this minimum value.

QUESTION 4

6 MARKS

a) **WITHOUT** solving the equation, determine the nature of roots of $10x^2 + x - 1 = 0$ (4)

b) Determine the nature of roots of the function graphed below. (2)



QUESTION 5

8 MARKS

Steve and Paul want to ride their bicycles from school to their rugby fixture. Steve leaves school 20 min after Paul. Paul is traveling 5 km/h slower than Steve. 1 hour after Steve leaves school he catches up to Paul. How fast was Steve travelling? Complete the table below to help you. (8)

	Distance	Speed	Time
Paul			
Steve		x	1 hour